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Keeping Tables Intact Under a Token Budget: Structure, Simple Baselines and Answer Format in Local Retrieval-Augmented Question Answering

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Abstract: Retrieval-augmented generation on local hardware must fit evidence into a small context, and pages whose answers sit in tables raise the question of how much of their structure to keep. We study it with Précis, an offline pipeline that keeps HTML tables and lists in native form and packs them first under an estimated-token ceiling with a guarded structural-priority rule (G), against cheap baselines on table-bearing Natural Questions pages. On short pages, the system's own rewrite of tables as extracted sentences cut exact match (EM) to 14.0%, against 49.5% for native tables (in-sample); against tag-stripped chunks, the cheap structure-keeping baselines were never significantly ahead (−2.5 to +6.0 points at the 2,048 ceiling on the protocol-first cohorts). On a fresh cohort G (48.0% EM) was level with raw-markup, table-aware and hybrid chunks and whole-document Markdown (47.0 to 49.0%), with three generators from two model families. On 200 full Wikipedia pages, where the 2,048-token ceiling binds, G was the worst arm (22.0% EM against 33.0 to 41.5%): same-document fact suppression removed every prose fact, leaving lists and reference text in half-filled contexts; a repaired rule (G2) raised EM on a fresh draw from 17.5% to 33.5%, non-inferior to table-aware chunking. A short-answer prompt format raised EM by 25.0 to 36.5 points, but a language-model judge, checked against two blind author labellers (92% and 97% agreement), saw gains of 1.5 to 12.0 points. Native structure should be kept but not at the expense of prose, and strict-match comparisons conflate answer format with accuracy.

Keywords: Answer-Format Sensitivity, Context Budget, Evaluation Methodology, Non-inferiority, Retrieval-Augmented Generation, Table Question Answering
Categories: H.3.3, I.2.7
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1 Introduction

Retrieval-augmented generation (RAG) systems retrieve text passages and place them in a language-model prompt [Lewis et al. 2020, Gao et al. 2024]. On local hardware, small generators and limited memory favour short prompts, and long contexts are used unevenly [Liu et al. 2024], so evidence has to be selected under a budget; this paper treats that budget as a design parameter (2,048 estimated tokens in most experiments), not as a measured hardware limit. Documents whose answers live in tables and lists make the selection harder: a table cell has meaning only through its row and column headers, and chunking, stripping markup or rewriting a table as sentences can sever those links.

This paper asks how much a structure-aware packing rule adds over simple baselines that also keep structure, and how much of a measured difference between such systems is an artefact of answer format. The testbed is Précis, an offline pipeline that keeps HTML tables and lists in native form, packs them first under an estimated-token ceiling with a guarded structural-priority rule (Candidate G), and fills any remaining budget with subject–verb–object–adjunct (SVOA) facts extracted from prose. No experiment isolates the fact layer's contribution inside G, to which it added only 111 facts over 200 contexts; where the fact-only configuration (Précis v3) was compared with plain chunks (Sections 6.6 and 6.7) it was behind, and an audit (Section 6.8) found its negation flag unreliable. Its specification is in the full manuscript in the repository.

Three research questions guide the evaluation. RQ1 (structure and baselines): for table- and list-bearing pages, how does native-structure packing compare with structure-destroying conversion and with cheap structure-keeping baselines, and what happens when the ceiling binds? RQ2 (answer format): how much of a strict-match difference between prompts comes from answer format and how much from the abstention instruction, and does an answer-equivalence measure change the conclusion? RQ3 (information loss): at which stage is answer-bearing evidence lost?

The contributions are evaluative: (i) an ablation showing that the system's own rewrite of tables as extracted sentences is the main source of loss on a table-bearing cohort (in-sample for G), and no significant advantage of the cheap structure-keeping baselines over tag-stripped chunks on the protocol-first cohorts; (ii) a protocol-first comparison on a fresh cohort against five simple baselines, with non-inferiority verdicts over margins of 3 to 7 points, table and list strata, exploratory ceiling sweeps over single documents and five-document corpora, and a protocol-first cohort of full Wikipedia pages on which the ceiling binds; (iii) a 2×2 decomposition of a concise-answer prompt into answer format and abstention wording, scored by exact match and by an answer-equivalence judge checked against human labels; (iv) a stage-loss trace on two cohorts; and (v) negative results on the fact layer: its fact-only contexts are behind plain chunks and its negation labels are unreliable. Extending [Sui et al. 2024] and HtmlRAG [Tan et al. 2025], the study sets a structure-first rule against table-aware, hybrid and whole-document baselines on the same table-bearing questions under one budget, including full pages, and separates answer format from abstention wording. The pipeline, protocols, per-case records and scripts are released (Data and Code Availability).

Section 2 reviews related work, Section 3 formulates the packing problem, Sections 4 and 5 describe the system, Section 6 reports results, discussion and limitations, and Section 7 concludes. Appendix A describes the prompt templates, and Appendix B points to the fact-layer specification and implementation notes in the full manuscript.

2 Related Work

Related work is grouped into seven themes: retrieval-augmented generation and its context limits, context compression, retrieval granularity, tables and HTML as evidence, epistemic status in text, hierarchical and hybrid retrieval, and answer evaluation and prompt sensitivity.

The original RAG formulation couples a parametric sequence-to-sequence model with a retrieved non-parametric memory [Lewis et al. 2020], and dense dual-encoder retrieval became its standard front end [Karpukhin et al. 2020]. A survey catalogues the retrieval, augmentation and generation variants [Gao et al. 2024]. Liu et al. show that accuracy degrades when relevant evidence sits in the middle of a long input [Liu et al. 2024], and Barnett et al. catalogue failure points of engineered RAG systems, among them chunking and retrieval choices [Barnett et al. 2024]. Self-RAG trains a model to decide when to retrieve and to critique its output [Asai et al. 2024]; such systems assume a model that can be fine-tuned or a window that is generous. Long-context models outperform RAG on average when well resourced, at a much higher cost [Li et al. 2024], retrieval helps even models with extended windows [Xu et al. 2024a], and adding retrieved passages to a long-context model can lower its accuracy [Jin et al. 2025]; whole-document input is therefore one of the baselines here.

Context compression reduces the prompt directly. LLMLingua prunes tokens using a small language model and a budget controller [Jiang et al. 2023]; Selective Context removes lexical units with low self-information [Li et al. 2023]; and RECOMP compresses retrieved documents into short summaries [Xu et al. 2024b]. All three need an additional neural model at inference time, and none treats a table as a unit to be kept or dropped whole.

A second family changes the unit of retrieval. Dense X Retrieval shows that indexing propositions rather than passages can improve retrieval and downstream question answering [Chen et al. 2024a], and FActScore uses atomic facts to evaluate generated text [Min et al. 2023]. GraphRAG builds an entity graph and community summaries for corpus-level questions [Edge et al. 2024], at the cost of many LLM calls during indexing. Element-based chunking, which splits a document at its titles, tables and list items, improved RAG on financial reports [Jimeno Yepes et al. 2024]; the table-aware baseline TC of Section 6.3 follows that idea. Open information extraction supplies classical machinery for tuples [Banko et al. 2007, Fader et al. 2011, Angeli et al. 2015, Kolluru et al. 2020].

Tables are a third kind of evidence. Table question answering is usually studied with semantic parsers or table-pretrained encoders [Pasupat and Liang 2015, Herzig et al. 2020], which presuppose a clean table; open-domain question answering over tables, and over tables and text, adds table retrieval [Herzig et al. 2021, Chen et al. 2021], and TableRAG retrieves schema and cell information so that a model can reason over very large tables [Chen et al. 2024b]. Sui et al. benchmark how the serialisation of a table in the prompt, including markup-based formats, changes what language models can answer [Sui et al. 2024], and HtmlRAG keeps cleaned and pruned HTML for retrieved web pages and, under a fixed context window, finds it better than plain-text and Markdown conversions and than chunk-based refiners on several question-answering sets, Natural Questions among them [Tan et al. 2025]. A general RAG pipeline receives HTML in which tables and prose are mixed, and what it does with a table before prompting decides what the generator can use. None of them compares a structure-first packing rule with table-aware and hybrid chunking and whole-document Markdown on the same table-bearing questions under one token budget, or examines such a rule on long pages

where that budget binds. Précis addresses that preprocessing and packing step rather than the table-reading model.

Finally, the epistemic status of a claim has a long history. FactBank annotates event factuality along source, polarity and certainty [Saurí and Pustejovsky 2009], NegEx popularised rule-based negation detection with trigger terms and scope windows [Chapman et al. 2001], and the ∗SEM 2012 shared task addressed negation scope and focus [Morante and Blanco 2012]. Précis keeps a small, closed set of modalities and rule-based negation because both are cheap, inspectable and runnable on a CPU.

Retrieval can also be hierarchical or hybrid. RAPTOR builds a tree of recursive summaries and retrieves across its levels [Sarthi et al. 2024]; Précis instead uses the document's own markup and no learned summaries. Combining lexical and dense rankings is a standard, strong and cheap baseline: BM25 [Robertson and Zaragoza 2009] fused with a dense ranking by reciprocal rank fusion [Cormack et al. 2009]; a cross-encoder reranker [Nogueira and Cho 2019] is a stronger and costlier second stage that is not evaluated here. Section 6.3 therefore compares G with table-aware chunking and a hybrid retriever, not only with plain chunking.

Evaluation choices matter as much as retrieval choices. Token-level metrics such as exact match undercount correct answers [Bulian et al. 2022], model-judged answer equivalence tracks human judgement more closely [Kamalloo et al. 2023], although language-model judges have known position and verbosity biases [Zheng et al. 2023], and language-model accuracy can move by tens of points with small changes of prompt format [Sclar et al. 2024]. RQ2 is positioned against this work.

3 Problem Formulation

A document is split into prose and structural elements. A structural element e is a table or list rendered in native form, with its own embedding and source document d(e). Each prose sentence yields zero or more facts t = (s, v, o, X, n, μ, a): subject, verb phrase, object, a set of typed adjuncts, a negation flag n, an epistemic modality $\mu \in$ {asserted, attributed, conditional, hypothetical} and, for attributed facts, the source a. Facts and elements are embedded into 384-dimensional unit vectors, and facts are stored in a set G.

Each fact carries an intrinsic score SDH(t) $\in$ [0, 1], an equally weighted sum of entity relatedness, verb informativeness, uniqueness and adjunct density (specified in the full manuscript in the repository); its constants were set by hand and were neither tuned nor ablated. At query time a fact is ranked by the product of this score and its relevance to the question:

$$\pi(t,q) = \mathrm{SDH}(t) \cdot \mathrm{sim}(t,q), \qquad \mathrm{sim}(t,q) = 1 - d_{\cos}(e_t, e_q) \tag{1}$$

The token cost of a fact is estimated from its rendered word count $\omega(t)$, including the negation marker, adjuncts, modality tags and the "Fact *N*:" label. The 1.3 factor is a word-count proxy and was not calibrated against the generator's own tokenizer:

$$\tau(t) = \max\big(3,\ \lfloor 1.3\,\omega(t) \rfloor + 2\big) \tag{2}$$

Selecting the fact context is a 0/1 knapsack problem over the candidates C, which the system solves greedily in descending order of π (not of the value density π/τ, which was not evaluated):

$$\max_{\mathscr{P} \subseteq C} \sum_{t \in \mathscr{P}} \pi(t,q) \quad \text{subject to} \quad \sum_{t \in \mathscr{P}} \tau(t) \le B \tag{3}$$

Structural priority (Candidate G). Let $D_{\mathscr{P}}$ be the source documents of the facts packed by Eq. (3) and d' the source document of the top-ranked packed fact. A structural element e is relevant if sim(e,q) ≥ 0.25, or if it shares a keyword with the question and sim(e,q) ≥ 0.15. It is eligible if $d(e) \in D_{\mathscr{P}}$, if no fact was packed, or if sim(e,q) ≥ 0.40. Eligible elements are ranked by similarity with a document-affinity bonus, and at most three are selected:

$$\begin{aligned}\sigma(e,q) = \mathrm{sim}(e,q)\,\big(1 + 0.15 \cdot \mathbf{1}[d(e) \in D_{\mathscr{P}}] \\ + 0.10 \cdot \mathbf{1}[d(e) = d']\big)\end{aligned} \tag{4}$$

Selected elements are packed first, within B. If at least one is packed, every fact whose source document contributed a packed element is suppressed, as is every fact whose subject and object both occur in the packed structural text, and the remaining budget is filled with the other facts. If nothing structural fits, the context falls back to the fact-only packing of Eq. (3). The relative context reduction of G with respect to the persistent baseline F is the descriptive quantity

$$\Delta = 1 - \frac{T_G}{T_F}, \qquad T_V = \frac{1}{N}\sum_{i=1}^{N} \tau_V(i) \tag{5}$$

where $\tau_V(i)$ is the estimated context tokens of variant V on case i. Problem statement. Given a document collection, a question q and a hardware profile that fixes B, construct the fact store G and the structural store offline, then return a context of at most B estimated tokens, built from packed structural elements and a packed subset of facts chosen greedily for a high $\Sigma\ \pi$(t,q) while excluding hypothetical facts, and render it as a grounded prompt for a local language model. The ceiling B is enforced on estimated tokens (Eq. (2)); the generator's own count runs about 1.24 times higher (Section 6.2), so B bounds the estimate and not the native prompt length.

4 System Model

Runtime profile ceilings (1,024–4,096 tokens) are distinct from the 2,048-token ceiling of most evaluations (Section 6.4 varies it).

Fig. 1 shows the data flow of Précis from ingestion to local generation. A browser interface lets a user upload documents, ask questions and inspect the packed context, the pipeline trace and the hardware profile. A FastAPI service exposes ingestion, query and inspection endpoints with token streaming, authentication, rate limiting, input limits, a basic prompt-injection filter and a rule-based intent router; none of these service features was evaluated.

The data layer is a persistent ChromaDB collection [Chroma 2026] with a cosine HNSW index [Malkov and Yashunin 2020] for facts, a second collection for structural evidence such as tables and lists, and a bounded in-memory session store. Generation uses a locally served model through Ollama [Ollama 2026]; no stage calls a cloud service. Facts are embedded with all-MiniLM-L6-v2 [Wang et al. 2020], a 384-dimensional distilled encoder in the Sentence-BERT family [Reimers and Gurevych 2019], computed from a canonical text that includes the negation marker, adjuncts and modality so that "does not deliver" and "delivers" do not collapse to the same vector. Every component is free and open source.

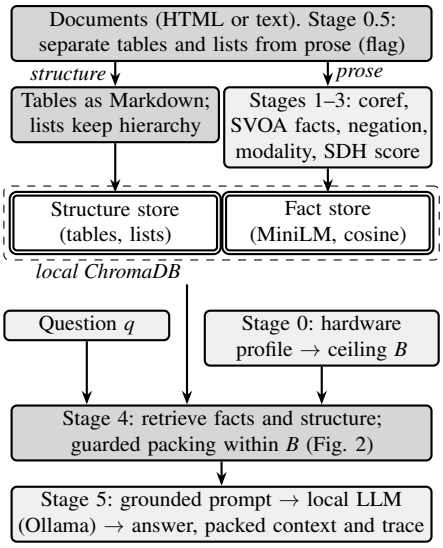


Figure 1: Précis system architecture: ingestion and extraction, local storage, similarity retrieval, Candidate G guarded context packing, hardware-budget control and local inference.

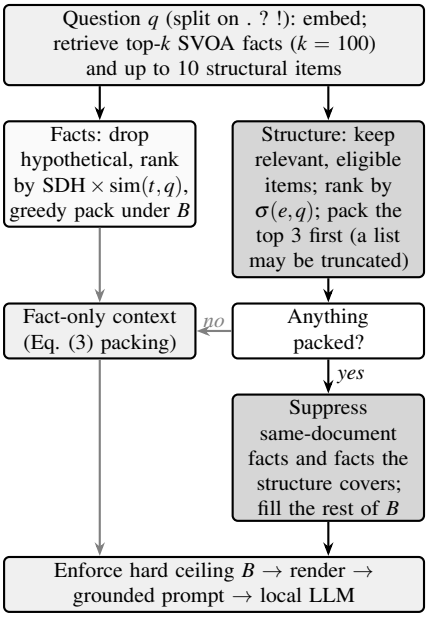


Figure 2: Query-time flow: fact packing (left) and guarded structural-priority packing (right) before the hard ceiling and generation.

5 Proposed Framework

The framework has an ingestion path, which runs once per document, and a query path, which runs per question (Fig. 2). A question costs one embedding, two nearest-neighbour searches (facts and structure), a sort and a greedy pass; extraction and scoring are not repeated.

Stage 0: Hardware profiling. The profiler reads CPU cores, RAM and GPU memory and selects a token ceiling of 4,096 (RAM $\geq$ 8 GB and VRAM $\geq$ 6 GB), 3,072 (RAM $\geq$ 8 GB and VRAM $\geq$ 2 GB), 2,048 (RAM $\geq$ 16 GB without a qualifying GPU) or 1,024 (other hosts), with a default model per mode (hardware.py). These rules were not evaluated: in the experiments the ceiling was fixed at 2,048 estimated tokens unless Section 6.4 varies it, and the generator was fixed per experiment. The evaluation host was an AMD Ryzen 9 8945HS laptop with 16 GB RAM and an NVIDIA RTX 4060 Laptop GPU (8 GB VRAM); CPU-only operation, memory use and ingestion and retrieval latency were not measured; generation time is reported in Section 6.9.

Stage 0.5: Structural retention. When the feature flag PRECIS_ENABLE_LIST_EXTRACTION is on, HTML tables and lists are separated from prose; the flag is off by default, so the released application answers with the fact-only configuration (Précis v3 below) unless it is set, and with G when it is. Tables are rendered as compact GitHub-flavoured Markdown and scored per question: 2 for each header matching a question word, 4 for each header matching a requested attribute, 3 for each caption word and 1 for any cell match. A table is selected at a score of at least 2, or if it is a small infobox

(three rows or fewer) with a header or caption match, and it is rejected if the question names it as an exclusion. Lists keep their hierarchy, navigation lists are discarded, and tables that duplicate lists are removed.

Stages 1–3: Fact layer. F-coref [Otmazgin et al. 2022] replaces pronouns with their referents, a spaCy dependency parse [Honnibal et al. 2020] yields SVOA facts with rule-based negation and modality labels, and the facts are scored with SDH and written to ChromaDB with their modality, adjuncts, attribution, source document and sentence index (specified in the full manuscript).

Stage 4: Query-adaptive packing. Multi-part questions are split on sentence punctuation, and up to $k = 100$ candidates per part are retrieved and merged. Hypothetical facts are removed by default; survivors are sorted by Eq. (1), with modality priority (asserted, attributed, conditional) breaking ties, and added greedily while the running cost stays within B. A fact that does not fit is skipped and later, cheaper facts may still be added. In Candidate G, up to ten structural items are retrieved in parallel and handled by Eq. (4) and the suppression rule of Section 3 (Fig. 2, right branch). Truncation applies only to a list that does not fit (it is cut at a line boundary if more than 30 tokens remain after the header), and the assembled context is trimmed to the hard ceiling as a last step.

Stage 5: Local generation. Packed content is inserted into a grounding prompt that tells the model to use only the supplied context, to preserve negation, and not to present attributed, conditional or hypothetical claims as established. The application supports a *standard* and a *concise* answer-formatting template; the controlled experiments below fix the template per condition.

The complete procedure is given as Algorithm 1 in the full manuscript in the repository.

6 Results and Discussion

6.1 Experimental Setup

All results come from the authors' harness on the evaluation host of Table 1, and the cohorts are analysed separately and never pooled.

Scoring is the same in every cohort. A prediction is correct for exact match if its normalised text equals the gold answer or any alternative answer that MRQA lists, and correct for containment if a normalised answer occurs in it at word boundaries. The rule is implemented once (scripts/scoring.py); re-scoring the prompt experiment, which was first scored against the single gold answer, changed 3 of 800 exact-match and 9 of 800 containment outcomes and left the primary test of Section 6.6 unchanged. ROUGE-L, MiniLM cosine and BERTScore, and the reference shown to the equivalence judge, use the single gold answer.

Inference plan. Each analysis has one declared status. (a) The original table cohort (Table 2) is in-sample, because G was designed on it, and descriptive (unadjusted exact McNemar tests, question-level bootstrap intervals). (b) On the fresh cohort the primary family (exact match; G against D, F2, N_raw and N_clean; two-sided exact McNemar, Holm over four contrasts) was fixed in a protocol before any model call. (c) Baseline adequacy (Section 6.3) compares G with seven arms by the lower bound of a 99% question-level bootstrap interval (10,000 resamples); because the 5-point non-inferiority margin was chosen after the fresh-cohort results (and before W, TC and H were run), verdicts are given for margins of 3 to 7 points, with Newcombe paired score intervals and a Bonferroni level for seven comparisons as checks. (d) In the prompt experiment

(Section 6.6) the primary test is one-sided McNemar for qwen3.5:4b with Précis v3. (e) On the long-document cohort (Section 6.4) the primary family (exact match at the 2,048 ceiling; G against N, N_raw, W, TC and H; non-inferiority at 5 points by 99% bootstrap intervals; two-sided exact McNemar with Holm) and the margin were fixed before the draw, and G2 was tested on a second, fresh draw under its own protocol (G2 against G, and non-inferiority to TC at 5 points by a 97.5% interval; Holm over the two). Everything else (the ceiling sweeps, with Holm over their 15 contrasts, the threshold sensitivity, the strata and the 2×2 follow-up) is exploratory. The protocols are author-held files whose SHA-256 hashes and time stamps were written before the model calls; they were neither registered externally nor under version control during the runs, so their order relative to the data cannot be verified independently, and they are called protocol-first below. Clustering is handled by resampling questions (187 distinct questions in the 200 cases of the original cohort, 74 in the 200-sample benchmark, distinct by construction elsewhere). Each case is a single greedy generation; repeating identical contexts changed 3 of 200 exact-match outcomes (Section 6.5), the size of the smallest differences reported, so verdicts that rest on one or two cases are within that noise. With 10 to 20% discordant pairs, 200 paired cases detect about 6 to 9 points at 80% power (two-sided 5% level). A non-significant difference is never read as equivalence.

Parameter	*Specification*
Data source	MRQA 2019 [Fisch et al. 2019] version of Natural Questions [Kwiatkowski et al. 2019], development split; full NQ v1.0 development pages for the long-document cohort
Inference	Local Ollama [Ollama 2026]; qwen2.5-coder:7b [Hui et al. 2024] and qwen3.5:4b [Qwen Team 2026a]; no cloud calls
Decoding	Temperature 0.0, top-p 0.9, seed 42, at most 512 generated tokens
Context ceiling	2,048 estimated tokens (word-count proxy, Eq. (2))
Additional baselines	W: whole-document Markdown, truncated at the ceiling; TC: table-aware chunks, dense retrieval; H: the same chunks, BM25 + dense with reciprocal rank fusion [Robertson and Zaragoza 2009, Cormack et al. 2009]
Fact store and encoder	Embedded ChromaDB, cosine distance; all-MiniLM-L6-v2 (384-d)
Chunked baseline	512-word chunks, 128-word overlap, top-5 retrieval (used as "Naive RAG" in Section 6.6)
Metrics	Exact match (EM) and word-boundary containment (gold answer or any MRQA alternative), ROUGE-L [Lin 2004], BERTScore [Zhang et al. 2020] (deberta-large-mnli [He et al. 2021]), MiniLM cosine
Statistics	Exact McNemar (paired binary), Holm / Bonferroni / Benjamini–Hochberg, question-level bootstrap (10,000 resamples), Newcombe paired score intervals, Wilson intervals
Evaluation host	AMD Ryzen 9 8945HS, 16 GB RAM, RTX 4060 Laptop GPU (8 GB VRAM); spaCy 3.8.14, en_core_web_lg 3.8.0
Replay generator and judge	qwen3.5:9b [Qwen Team 2026b] and gemma4:e4b [Google DeepMind 2026] (thinking off) on the stored, generator-independent contexts; gemma4:12b [Google DeepMind 2026] (Ollama) as answer-equivalence judge, temperature 0

Table 1: Experimental configuration of the controlled evaluations.

Use of generative AI tools. During the preparation of this manuscript the authors used ChatGPT (OpenAI, GPT-5) and Claude (Anthropic, Claude Opus 5.5 and Claude Sonnet 5.5) for language editing, organisation of existing material and review of analytical summaries. Claude Code (Anthropic; Claude Opus 5.5 and Claude Sonnet 5.5) wrote

analysis scripts, ran the experiments of Sections 6.3–6.6, the third-generator runs, the long-document and repaired-G cohorts and the regeneration checks of Section 6.10, re-scored every cohort under one scoring rule, computed the re-analyses of the later versions (scripts/paper_checks_v4.py, scripts/paper_checks_v5.py, the experiment analysis scripts and scripts/revise_paper_v8.py), drew Figs. 1 to 4 (scripts/plot_figures_results.py and the TikZ sources in paper/venues/common), and revised the manuscript in response to simulated peer reviews produced with Claude-based tools; those reviews are not independent of the model that revised the text. A Claude-based subagent acted as the second rater of the pilot error audit (Section 6.6) and saw only the question, gold answer, evidence context and the two unlabelled outputs. A local model (gemma4:12b) was the answer-equivalence judge, and a language model reworded the free-text notes of one annotator in the extractor audit; all other labels were assigned by human annotators. Every reported number is produced by a script in the repository from the stored per-case records. The authors are responsible for verifying the data, statistical calculations, references and interpretation, and all authors reviewed and approved the final manuscript.

6.2 Structural Preservation on Table-Bearing Documents (RQ1)

The table-bearing cohort has 200 cases drawn with seed 42 from MRQA Natural Questions development documents whose raw HTML contains a table, unordered-list or ordered-list marker, after excluding 600 historically used identifiers. The 200 cases hold 187 distinct questions (12 questions occur more than once), so the paired tests treat a few repeated questions as independent. The gold answer string occurs in the text of at least one table for 164 of the 200 cases (82%) and elsewhere in the source document for the other 36 (a string-presence check); cases were selected for table or list markup, not for answers located in a table. The same draw also produced a 100-case prose-only stratum, which was never run. All variants used qwen2.5-coder:7b with greedy decoding, the concise prompt and a 2,048-token ceiling. Predictions for variants A–F were reused unchanged from frozen records, and G was generated live. The variants are: A, a prose baseline (raw document HTML passed through coreference resolution and prose-only SVOA extraction, with no table or list handling and no passage chunking); B, SVOA-decompiled prose, in which the system's own extractor rewrites tables as sentences; C, conservative table retention (distance threshold 2.0); D, table retention alone (threshold 0.0, tables kept as Markdown, no lists or facts); E, an in-memory prototype retaining tables and lists; F, the persistent production pipeline (ChromaDB storage, similarity-dominant ranking, residual context filled with SVOA facts); and G, Candidate G, which adds guarded structural-priority packing.

Var.	*Pipeline*	*Exact match*	*Containment*	*G − var. EM (pp)*	*p, EM*	*G − var. Cont. (pp)*	*p, Cont.*
A	Prose baseline	64/200 (32.0%)	87/200 (43.5%)	+23.0	1.7×10^{-9}	+25.0	1.5×10^{-9}
B	SVOA-decompiled prose	28/200 (14.0%)	39/200 (19.5%)	—	—	—	—
C	Table retention, $\theta = 2.0$	61/200 (30.5%)	79/200 (39.5%)	—	—	—	—
D	Table retention alone, $\theta = 0.0$	99/200 (49.5%)	122/200 (61.0%)	+5.5	0.019	+7.5	0.0015
E	Prototype tables + lists	108/200 (54.0%)	137/200 (68.5%)	+1.0	0.727	0.0	1.000
F	Persistent production pipeline	105/200 (52.5%)	131/200 (65.5%)	+2.5	0.424	+3.0	0.307
G	Guarded structural priority	110/200 (55.0%)	137/200 (68.5%)	ref.	ref.	ref.	ref.

Table 2: Structural-preservation ablation on the table-bearing cohort (N = 200). p-values are unadjusted two-sided exact McNemar tests.

Table 2 gives the results. Rewriting tables as extracted sentences (B, 14.0% EM) was the weakest variant and fell below the prose baseline (A, 32.0%), whereas retaining tables in native form (D) reached 49.5%; D beat B in 79 cases against 8 (exact McNemar $p = 8 \times 10^{-16}$; question-level 95% interval +27.3 to +43.9 points). B measures the cost of this extractor's conversion, which has the defects reported in Section 6.8, not the cost of every tables-to-text method. Candidate G had the highest exact match (110/200, 55.0%) and, with E, the highest containment (137/200). Its leads over F (+2.5 points, 15 against 10 discordant pairs, p = 0.42; 95% interval −2.5 to +7.7) and E (+1.0, p = 0.73) are not distinguishable from zero, and its lead over D (+5.5, p = 0.019) does not survive Holm or Bonferroni correction over the four G contrasts (0.058 and 0.077). Because G was designed on this cohort, these results are descriptive.

Fresh cohort. A second table- or list-bearing cohort was drawn under a protocol-first design: 200 distinct questions from the Natural Questions development file whose documents contain table or list markup, excluding every identifier and question used before (1,425 candidate records, 748 distinct questions; seed 42). Five arms were run with qwen2.5-coder:7b, the concise prompt and the 2,048 ceiling: chunked passage retrieval on the raw document (N_raw) and on tag-stripped text (N_clean), table retention alone (D), a reconstruction of F without same-document fact suppression (F2; the frozen F cannot be regenerated by the current code) and G. In the primary family G exceeded D (20 against 3 discordant cases, Holm p = 0.002) but not F2 (13 against 6, Holm p = 0.50), N_raw (11 against 9, p = 0.82) or N_clean (18 against 10, p = 0.50). The lead over D is confined to the 32 list-only documents, where D has no evidence by construction (G 18 correct, D 0), so it is a mechanism check. On the 168 documents with a table, G, D and N_raw (and W, TC and H of Section 6.3) scored 76 to 79 correct, against 71 for tag-stripped chunks and for F2; native tables beat stripped chunks (D against N_clean) by 4.8 points (17 against 9, p = 0.17, 95% interval −1.2 to +10.7), which has the expected sign but is not significant at this size. The large structure effect of Table 2 is therefore a loss from this conversion, not a gain over cheap structure-keeping alternatives. Taking every cheap structure-keeping baseline (N_raw, W, TC and H) that was run on a protocol-first cohort, at the 2,048 ceiling and against that cohort's tag-stripped chunks, gives ten comparisons: +3.0 to +5.0 points on the fresh cohort, and −2.5 (N_raw) to +6.0 (TC) on the two long-document draws; none is significant (smallest p = 0.076). At 1,024 tokens on long pages raw-markup chunks were behind (−10.0 points, p = 0.001), and D, which keeps tables but not lists, was behind on the whole fresh cohort (−4.5). In the exploratory sweeps (Section 6.4) H and W against budgeted stripped chunks ranged from −5.5 to +8.5 points, none significant after Holm correction over these ten (smallest unadjusted p = 0.0095). Keeping structure was therefore never shown to beat stripping it at the main ceiling, and keeping raw markup can cost accuracy on long pages.

Context size. Against the authors' own pipelines G's packed context was smaller: 56.2% fewer estimated tokens than F2 on the fresh cohort (401.8 against 917.9; paired mean difference −516 tokens, 95% interval −573 to −461) and 53.2% fewer than F on the original cohort (365.8 against 781.1), or 43 to 46% fewer model-reported prompt tokens. Against external baselines it is not smaller (Section 6.3). The estimate understates native length: native prompt tokens ran at about 1.24 per estimated context token, and at the 2,048 ceiling the full native prompt exceeded 2,048 tokens in 41 F cases and 1 G case.

6.3 Baseline Adequacy and Efficiency (RQ1)

The fresh cohort was re-run with three further arms that do not use the SVOA or SDH code and are cheap to build: W, the whole document converted from HTML to

Markdown in document order (tables as Markdown tables, lists as bullets) and truncated at the ceiling; TC, table-aware chunks (tables kept whole or split by rows with the header repeated, lists kept whole, prose grouped to about 128 words) retrieved by dense similarity; and H, the same chunks retrieved by BM25 and dense rankings fused by reciprocal rank fusion. All use the 2,048-token ceiling, at most five chunks, qwen2.5-coder:7b and the concise prompt (protocol and hash in the repository). Table 3 gives exact match, containment, mean estimated context tokens, exact-match points per 100 tokens, the exact-match difference of G from each arm with its 99% interval, and the smallest margin, among 3 to 7 points, at which G is non-inferior. The last column gives exact match when the same contexts are answered by a general-purpose generator, qwen3.5:9b, because qwen2.5-coder is a code model.

Arm	*EM /200*	*Cont. /200*	*Mean tokens*	*EM pp per 100 tok*	*G − arm, EM pp [99% CI]*	*Smallest NI margin met (pp)*	*EM, qwen3.5:9b*
N_raw (chunks, markup)	94	124	503	9.3	+1.0 [-4.5, 7.0]	5	99
N_clean (chunks, stripped)	88	115	280	15.7	+4.0 [-2.5, 11.0]	3	106
W (whole doc, Markdown)	98	129	388	12.6	-1.0 [-4.5, 2.5]	5	100
TC (table-aware chunks)	96	126	420	11.4	+0.0 [-4.5, 5.0]	5	98
H (hybrid, table-aware)	98	128	420	11.7	-1.0 [-6.0, 4.0]	7	98
D (tables only)	79	104	351	11.3	+8.5 [3.0, 14.5]	G higher	84
F2 (F without suppression)	89	120	918	4.8	+3.5 [-2.0, 9.5]	3	85
G (guarded priority)	96	127	402	11.9	reference	—	98

Table 3: Baseline adequacy on the fresh table- and list-bearing cohort (N = 200, 2,048-token ceiling, qwen2.5-coder:7b except the last column). N_raw, N_clean, D, F2 and G are the earlier records; W, TC and H were run for this table. G − arm is positive when G is ahead. The margin column uses the 99% bootstrap lower bound; Newcombe paired intervals give the same values except 6 for H.

G was not distinguishable from any structure-keeping baseline: its exact-match differences from N_raw, W, TC and H lay between −1.0 and +1.0 points. Whether this counts as non-inferiority depends on a margin chosen after the fresh-cohort results were known. At 5 points G is non-inferior to N_raw, N_clean, W, TC and F2 but not to H; at 4 points only N_clean and F2 remain; and at a Bonferroni level for seven comparisons (99.3%) the Newcombe lower bounds against N_raw and TC (−5.1) cross the 5-point margin and that against W (−4.95) just meets it. The verdicts against N_raw, W and TC therefore rest on one or two cases, and non-inferiority to TC with its absence for H is not itself a supported difference. Efficiency does not favour G either: G was third of eight in exact match per 100 estimated tokens (Table 3), and its context was 13.5 tokens larger than W's (paired 95% interval +1.0 to +29.9), about 18 tokens smaller than TC's and H's, 102 smaller than N_raw's and 122 larger than N_clean's. These counts cover the context only: G also needs coreference resolution, parsing and SDH scoring at ingestion, whereas W and TC need only an HTML conversion.

Retrieval and packing make no model call, so each arm's context is the same whichever generator answers it, and answering the stored contexts with another generator is equivalent to running the whole pipeline with it. With qwen3.5:9b, G was +7.0 points above D ($p = 0.004$) and +6.5 above F2 ($p = 0.004$), and was not distinguishable from

N_raw (−0.5, p = 1.00), N_clean (−4.0, p = 0.18), W (−1.0), TC (+0.0) or H (+0.0). A third generator from another family, gemma4:e4b (exploratory, protocol-first), had lower exact match throughout (78 for G) and put G in the same position relative to the structure-keeping baselines: not distinguishable from N_raw (+1.0), N_clean (+1.0), W (−2.0, p = 0.22), TC (−2.5, p = 0.27) or H (0.0), and above D (+6.5, p = 0.002); only its lead over F2 was not reproduced (−1.5, p = 0.58). The position of G relative to the baselines therefore does not depend on the code-specialised generator.

6.4 Where the Ceiling Binds (RQ1)

At the 2,048 ceiling the budget barely binds on this data: G's largest context was 1,163 estimated tokens, and in the 200-sample benchmark of Section 6.7 186 of 200 passages fit in one 512-word chunk. MRQA documents are short (the longest in the 12,836-record development file has 800 words), which is consistent with the release truncating long pages. Two exploratory experiments therefore make the ceiling bind, and a third, protocol-first cohort uses full pages (below). First, the same single documents were run with ceilings of 256, 512 and 1,024 estimated tokens, comparing G with budgeted chunks (the N arm, tag-stripped 512-word chunks cut at the ceiling), H and W. Second, each question was placed in a five-document corpus (the relevant document and four random distractors, in shuffled order; mean 1,523 estimated tokens) with ceilings of 1,024 and 2,048; G ingested all five documents, the chunked arms treated the concatenation as one text, and W truncated it. Fig. 3 gives exact match; the full manuscript in the repository also tabulates the number of the 200 contexts that contain an answer string.

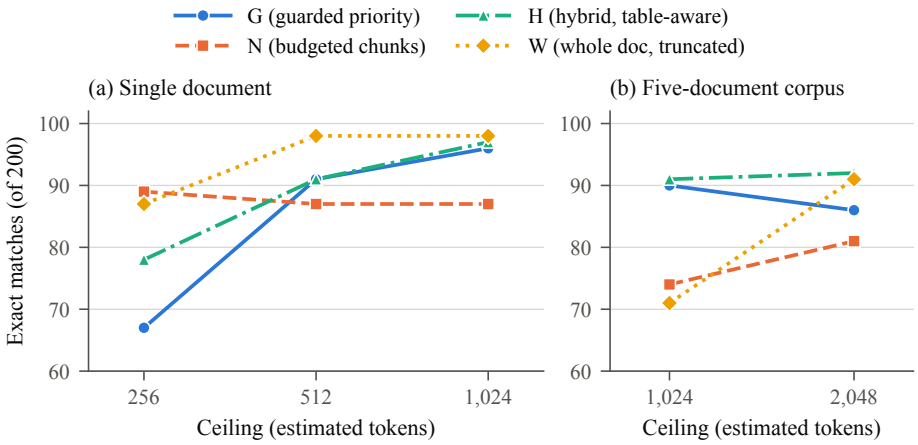


Figure 3: Exploratory ceiling sweeps (N = 200, qwen2.5-coder:7b, concise prompt): exact match against the ceiling for G, budgeted tag-stripped chunks (N), hybrid table-aware retrieval (H) and truncated whole-document Markdown (W), on (a) single documents and (b) five-document corpora.

With a single document and a 256-token ceiling the budget binds and G loses: exact match 67 against 89 for chunks (−11.0 points, 11 against 33 discordant cases, Holm p =

0.018 over the 15 contrasts of Fig. 3) and 87 for W (Holm p = 0.008), with only 150 of 200 G contexts still containing an answer string against 191 for chunks. The likely cause is that G packs a structural element first and truncates one that does not fit, which was not examined separately; row-level truncation with repeated headers, as in TC, is the obvious remedy. At 512 tokens and above no contrast survived Holm correction (smallest: G against truncated Markdown in five-document corpora at 1,024, 90 against 71, Holm p = 0.056), and G was never ahead of H (Fig. 3). The sweeps show a clear disadvantage of G at a very small ceiling and no advantage over hybrid table-aware retrieval in any setting.

What the ceiling bounds. B applies to the word-count estimate of Eq. (2), not to the generator's tokens; the generator was given a context window large enough for every prompt, so nothing was truncated by the model server.

Long documents. Because MRQA cuts every page to at most 800 words, a third cohort was drawn under a protocol fixed before the draw, with the primary family and a 5-point non-inferiority margin stated in advance: 200 questions from the full development pages of Natural Questions v1.0 [Kwiatkowski et al. 2019] whose page contains a table and at least 3,000 words of text, excluding every question used before (902 eligible questions among the 3,357 examples of the first three of the seven development shards, as the protocol fixed; seed 42). The median page had 7,093 words (9,625 estimated tokens), so the 2,048 ceiling covered about a fifth of it; the answer lay inside a table for 33 questions and beyond the first 1,575 words, which whole-document truncation can reach, for 57. Arms G, N, N_raw, W, TC and H were run as before with qwen2.5-coder:7b and the concise prompt at ceilings of 2,048 (primary) and 1,024 estimated tokens. Table 4 gives the results. A GPU out-of-memory error stopped the run once; a retry was added, the completed cases were kept and none was regenerated (as for the other interrupted runs, logged in the repository).

Arm (long documents)	*EM /200 at 2,048*	*Answer in context*	*Mean tokens*	*G − arm, EM pp [99% CI]*	*Holm p*	*EM /200 at 1,024*
G (guarded priority)	44	84	1,049	reference	—	43
N (chunks, stripped)	71	172	2,048	−13.5 [−23.0, −4.0]	0.001	75
N_raw (chunks, markup)	66	140	2,048	−11.0 [−20.0, −2.5]	0.002	55
W (whole doc, truncated)	75	168	2,048	−15.5 [−24.5, −6.5]	< 0.001	78
TC (table-aware chunks)	83	158	989	−19.5 [−28.5, −10.5]	< 0.001	79
H (hybrid, table-aware)	76	152	1,141	−16.0 [−25.0, −7.0]	< 0.001	76

Table 4: Long-document cohort (N = 200 full Natural Questions pages, median 7,093 words): exact match, contexts containing an answer string and mean estimated tokens at the 2,048 ceiling, the exact-match difference of G from each arm with its 99% bootstrap interval and Holm-adjusted McNemar p, and exact match at 1,024. qwen2.5-coder:7b, concise prompt.

When the ceiling bound on long pages, G was the worst arm, 11.0 to 19.5 points behind each of the five baselines (Holm $p \leq 0.002$; Table 4), so non-inferiority failed against all five at the pre-specified margin. At 1,024 the picture was the same (43 against 55 to 79), except that the gap to raw-markup chunks was not significant (−6.0 points, p = 0.10). Table-aware chunks were best (83 at 2,048, with a smaller context than G's). The loss arose before generation: an answer string was in only 84 of G's 200 contexts, against 140 to 172 for the baselines, and G was behind even on the 33 questions whose answer lay in a table (7 correct against 9 to 14). Fig. 4 sets these differences beside those of the fresh cohort (Table 3): there every interval except that against D includes zero,

here every interval lies below zero.

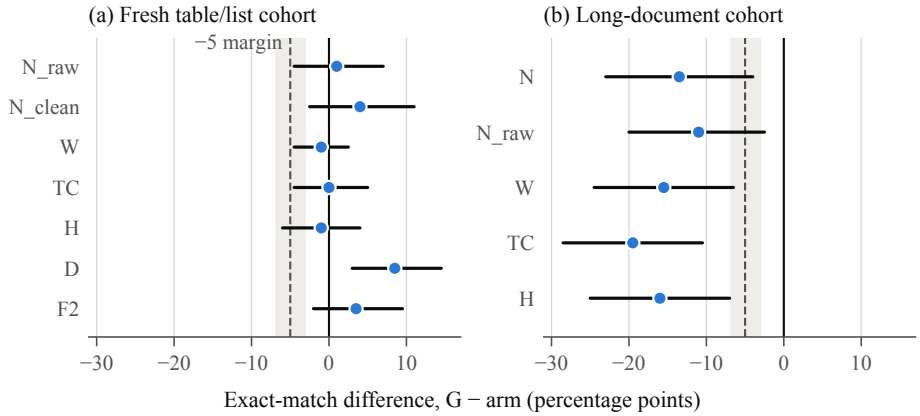


Figure 4: Exact-match difference of G from each arm (percentage points, positive when G is ahead) with its 99% question-level bootstrap interval, on (a) the fresh table- and list-bearing cohort (Table 3) and (b) the long-document cohort at the 2,048 ceiling (Table 4). The dashed line marks the 5-point non-inferiority margin and the shaded band the 3 to 7 points over which the fresh-cohort margin was varied.

Why G fails on full pages. A descriptive look at G's contexts, made after the results were known, shows the mechanism. None of G's 200 contexts contained a prose fact: on a single document, packing any table or list suppresses every fact of that document (Section 3), and full pages offer many lists. 82 contexts held lists only, and 107 contained reference-list, category or media-link text that the navigation filter, designed on MRQA's extracts, did not remove; G also left about half of the budget unused (1,049 estimated tokens on average). Same-document suppression and unfiltered reference lists, rather than the order of packing as such, are the likely reasons (a reading made after the results) that G is unsuitable for long single documents, and table-aware chunking keeps the benefit of native tables without that failure. The replays through other generators that the protocol listed as exploratory were not run, because the deficit lies in what the contexts contain, which does not depend on the generator.

Repair on a fresh draw. The diagnostic points to two changes, which were tested on questions not used to design them, under a protocol written before a second draw of 200 eligible full pages (none shared with the first; seed 43). The repaired rule G2 suppresses only facts whose subject and object both occur in the packed structural text, not every fact of the page, and skips a structural candidate when at least 30% of its lines are references or media and external links. On this draw G2 had 67 exact matches against 35 for G (+16.0 points, 38 against 6 discordant cases, Holm $p < 0.001$, 95% interval +10.0 to +22.5) and was non-inferior to table-aware chunks at the 5-point margin (63; +2.0 points, 97.5% interval −4.5 to +9.0); it was level with whole-document Markdown (69; −1.0, $p = 0.88$) and stripped chunks (61; +3.0, $p = 0.45$). Every G2 context contained prose facts (none of G's did), an answer string reached 158 of its contexts against 70 for G, and G2 used nearly the whole budget (1,958 estimated tokens on average, against 939 for table-aware chunks, so its parity comes at about twice their context). Table-aware

chunks scored 63 on this draw against 83 on the first, so differences between draws are as large as the margin. G2 was designed after the first long-document results and is tested on one further draw from the same source, so it is a repaired variant, not a separately tuned method; it still packs structure first, its two changes were not tested separately, and it is implemented in the experiment script (scripts/run_long_doc_fixed_g.py), not yet in the released pipeline.

6.5 Threshold Sensitivity of Candidate G

The thresholds of Eq. (4) and the cap of three elements were set on the original cohort. Perturbing each in turn on the fresh cohort (relevance 0.20 and 0.30, keyword-assisted 0.10 and 0.20, eligibility 0.30 and 0.50, cap two and four) changed 0 to 8 of 200 contexts and moved exact match by at most 1 point from the rerun of G (95 correct; the stored G has 96, and the rerun differed from it in 3 of 200 outcomes on identical contexts, which is the decoding noise of the local server at temperature 0). Because so few contexts change, this shows how rarely the cohort exercises the thresholds, not that they carry over to other data.

6.6 Answer Format and Abstention Policy (RQ2)

The main prompt experiment used 100 new question–document pairs from the same MRQA source, selected deterministically from an eligible pool of 3,904 by sorting SHA-256 hashes of normalised questions after removing overlaps with earlier cohorts. It crossed two generators (qwen2.5-coder:7b and qwen3.5:4b), two retrieval conditions (the chunked baseline, here "Naive RAG", and Précis v3, the fact-only pipeline without the structural packing of G) and two prompts (standard and concise), giving 800 evaluations on contexts cached once per question and condition and checked by SHA-256 parity. Hypotheses, guardrails and tests were written into protocol files time-stamped before the first model output; the guardrails were fixed after an earlier exploratory prompt comparison (on 101 and 100 other samples, not counted as evidence) had shown 48% containment. Runs used Ollama 0.35.1, Q4_K_M quantisation and the templates of Appendix A. The primary hypothesis was that the concise prompt raises Précis exact match for qwen3.5:4b (one-sided exact McNemar, $\alpha = 0.05$), with guardrails of a containment change of at least -0.25 and concise containment of at least 0.40; three secondary prompt effects used Holm-corrected one-sided tests.

Model / retrieval	*EM (std → conc.)*	*Containment*	*ROUGE-L*	*MiniLM*	*BERTScore*	*Mean tokens*
qwen2.5-coder:7b / Naive RAG	21 → 45	0.68 → 0.51	0.487 → 0.566	0.616 → 0.668	0.668 → 0.703	21.0 → 12.2
qwen2.5-coder:7b / Précis v3	18 → 35	0.53 → 0.41	0.416 → 0.445	0.565 → 0.553	0.632 → 0.635	20.7 → 7.4
qwen3.5:4b / Naive RAG	11 → 51	0.75 → 0.66	0.384 → 0.684	0.560 → 0.791	0.587 → 0.784	36.3 → 10.3
qwen3.5:4b / Précis v3	6 → 43	0.61 → 0.53	0.277 → 0.604	0.491 → 0.732	0.535 → 0.747	40.8 → 9.9

Table 5: Standard-to-concise prompt outcomes in the 2×2×2 prompt experiment (N = 100 per cell; EM in correct answers out of 100). EM and containment use the gold answer or any MRQA alternative; ROUGE-L, MiniLM and BERTScore use the single gold answer.

The primary hypothesis was supported: for qwen3.5:4b with Précis v3 exact match rose from 6/100 to 43/100 (37 concise-only against 0 standard-only correct answers; one-sided $p = 7.3 \times 10^{-12}$, identical under single-gold and alias scoring), and containment moved from 0.61 to 0.53, within both guardrails. The three secondary effects were also significant after Holm correction (11 to 51, 18 to 35 and 21 to 45 correct). Containment fell in all four conditions. A gain this large under strict matching is expected when one prompt asks for a bare span [Bulian et al. 2022, Sclar et al. 2024]; the follow-up below asks what it consists of.

The retrieval-by-prompt interaction was −0.03 (95% interval −0.13 to +0.07) for qwen3.5:4b and −0.07 (−0.15 to +0.01) for qwen2.5-coder:7b, so the prompt effect is not shown to differ between retrieval methods. The chunked baseline was numerically ahead of Précis v3, the fact-only configuration, in exact match and in containment in all four model–prompt cells of Table 5 (for example 51 against 43 correct for qwen3.5:4b under the concise prompt), in the same direction as the benchmark of Section 6.7; in exploratory paired tests it was ahead in containment in all four model–prompt strata (unadjusted p = 0.004 to 0.052), but after Holm correction over the eight tests only one difference remains below 0.05. Context coverage accounts for part of the gap: an answer string occurred in all 100 chunked contexts but in only 79 of the 100 fact-only contexts, so the fact layer lost the answer before generation in 21 cases.

Separating format from abstention. The concise prompt differs from the standard one in its answer-format instructions (a short span instead of a sentence) and in its abstention line (output "Insufficient information" instead of saying so explicitly). In a blinded 30-question pilot error audit (60 outputs labelled by one author and one language-model rater; $\kappa = 0.70$ on merged labels), the concise prompt corrected 12 answers and degraded none; 7 corrections unwrapped a correct entity from a sentence and 5 replaced an abstention. An exploratory follow-up crossed the two factors on the same 100 cached contexts by adding two prompts, each built by swapping only the abstention line (Appendix A): the concise prompt with the standard abstention line (short + explicit) and the standard prompt with the "Insufficient information" line (sentence + token). Table 6 gives, per cell, exact match, outputs judged equivalent to the gold answer by a local model (gemma4:12b, fixed rubric, temperature 0) and abstaining outputs.

Model / retrieval	*Sentence + explicit (standard)*	*Sentence + token*	*Short + explicit*	*Short + token (concise)*
qwen2.5-coder:7b / Naive RAG	21 / 84 / 2	24 / 56 / 36	55 / 79 / 4	45 / 64 / 22
qwen2.5-coder:7b / Précis v3	18 / 73 / 9	10 / 32 / 60	43 / 73 / 6	35 / 56 / 34
qwen3.5:4b / Naive RAG	11 / 77 / 17	18 / 74 / 19	51 / 81 / 3	51 / 78 / 9
qwen3.5:4b / Précis v3	6 / 56 / 30	17 / 59 / 30	43 / 70 / 8	43 / 67 / 14

Table 6: The 2×2 follow-up (N = 100 per cell): exact match / outputs judged equivalent to the gold answer / abstaining outputs. The judge agreed with two human labellers (authors, labelling blind) on 92% and 97% of the outputs each could label (Section 6.6).

Averaged over the abstention wording, the short-span format raised exact match by 25.0 to 36.5 points in the four model–retrieval combinations (95% intervals all above +18), so the large exact-match effect of the concise prompt is mainly a format effect. The abstention wording is a separate factor whose effect depends on the generator and the format. For qwen2.5-coder the "Insufficient information" line raised abstention (from 2 to 36 sentence outputs with Naive RAG and from 9 to 60 with Précis v3), lowered

judged-equivalent answers by 21.5 and 29.0 points on average, and lowered exact match in the short format (55 to 45 and 43 to 35; p = 0.006 and 0.021). These abstentions were mostly false: every abstention on a chunked context, and 46 of the 60 sentence-format abstentions on Précis v3 contexts, came from a context that contained an answer string. For qwen3.5:4b it left judged-equivalent answers almost unchanged (−3.0 and 0.0 points) and raised exact match in the sentence format (11 to 18 and 6 to 17; p = 0.016 and 0.001) but not in the short format, where both cells were unchanged; the format-by-abstention interaction for exact match was −11.0 points (95% interval −18 to −5) for qwen3.5:4b with Précis v3 and −13.0 (−22 to −4) for qwen2.5-coder with Naive RAG. The concise prompt of the main experiment therefore combines a format instruction that changes strict-match scores with an abstention instruction that, for the code-specialised generator, causes abstentions on answerable contexts.

Does the format effect survive an equivalence measure? Mostly not. Averaged over the abstention wording, the format effect on judged-equivalent answers was +1.5 and +4.0 points with Naive RAG (intervals including zero) and +12.0 and +11.0 with Précis v3 (95% intervals +6.5 to +18.0 and +3.5 to +18.5), so a short answer helped the judged score mainly on fact contexts, whose sentence fragments a sentence answer tends to restate. The headline cell, qwen3.5:4b with Précis v3, moved from 6 to 43 under exact match but from 56 to 67 under the judge. Exact match mostly measures whether an answer is a bare span, and many sentence answers of the standard prompt are equivalent to the gold answer. The retrieval ranking is the same under both measures: the chunked baseline was ahead of Précis v3 in all eight cells under exact match and by 6 to 24 judged-equivalent answers. The judge was checked by one author, who labelled 100 outputs sampled equally from its CORRECT and INCORRECT verdicts without seeing the verdict; 10 marked hard to tell were excluded. On the remaining 90 the two agreed on 83 (92.2%, Wilson 95% interval 85–96%; Cohen's $\kappa = 0.84$, which the verdict-balanced sample pushes upwards; 92.6% when the two verdicts are weighted by their frequency among the 1,600 outputs), on 46 of 49 sentence-format and 37 of 41 short-format outputs, so agreement does not differ visibly between the formats being compared, which is where a verbosity bias of the judge [Zheng et al. 2023] would show. The judge saw the single gold answer whereas exact match accepts MRQA alternatives, which can only make the judge stricter. A second author, who saw neither the first labels nor the verdicts, labelled the same 100 outputs (14 marked hard to tell): the judge agreed with this labeller on 83 of 86 (96.5%, Wilson 95% interval 90–99%; $\kappa = 0.93$; 97.6% weighted), and the two labellers agreed on all 76 outputs that both labelled. With two labellers and at most 90 scored outputs each, the judged counts are a supporting measure, and the conclusions about format rest on the direction of the differences, not their size.

6.7 Information Loss across Pipeline Stages (RQ3)

Single-passage benchmark. On a 200-sample development benchmark of short passages (74 distinct questions; 512-word chunks with 128-word overlap, standard prompt, qwen2.5-coder:7b, 2,048 ceiling), chunked retrieval was numerically ahead of the fact-only Précis v3: 57 against 38 exact matches (26 against 7 discordant pairs; question-level 95% interval for the difference −0.5 to +19.5 points) and 159 against 141 containment. 186 of the 200 passages fit in one chunk, so this setting tests extraction rather than packing, and it does not bear on the table cohorts.

The extraction rules were revised over successive versions (v1, v1.5, v2 and v3) after inspecting failures on this split, raising answer coverage (the answer string in some extracted fact) from 68.5% (v1) to 88.5% (v3) and exact match from 14.5% to 19.0%;

these are in-sample engineering results. Under one procedure without coreference, the current extractor covered 91.0% of these development documents and 86.6% of 1,000 held-out documents from the same file (95% interval 84.4–88.7%), so the level of coverage, though not the size of the improvement, carries over.

Tracing the answer string through the v3 pipeline gave 200/200 in the source passages, 177/200 in the extracted facts and 177/177 in both the retrieved candidates and the packed context, so in this setting the loss arose at extraction (23 cases) and after packing. Of the 177 covered cases, 141 outputs contained an answer string and 38 were exact matches, so at least 36 covered cases were lost at generation and 103 between containment and exact match. The 177/177 figures are conditional on extraction, and an answer string in a context shows that the answer was available, not that it was used.

The same trace on the table cohort is where retrieval and packing could lose evidence. The full manuscript in the repository tabulates it per arm. At the 2,048 ceiling the answer string was in the context in 197 of 200 cases for G and in 198 to 199 for the chunked, Markdown and F2 arms (168 for D, which has no evidence for the 32 list-only documents by construction), so the answer string survived retrieval and packing in nearly every context, and the remaining loss was at generation (abstentions and wrong answers) and at exact matching; 37 of G's 38 abstentions came from a context that held an answer string. The situation changes when the ceiling binds: at 256 estimated tokens the answer string was in only 150 of G's 200 contexts against 191 for chunks (Section 6.4), and that is where packing, not generation, decides the outcome. On full pages the same holds at 2,048: an answer string reached 84 of G's contexts against 140 to 172 for the other arms (Table 4).

6.8 The Fact Layer: an Extractor Audit

No experiment above isolates the SVOA fact layer's contribution to G. In the 200 contexts of G on the original cohort only 111 facts were packed in total, none with a negation marker, so in these cohorts it contributes little to what G packs; where it formed the whole context (Précis v3, Sections 6.6 and 6.7) it was behind plain chunks. What can be said is how accurate its labels are. Two human annotators, working independently, labelled, from 700 randomly drawn development documents (7,824 extracted facts, of which 184, 2.4%, were flagged negated), 400 facts sampled at random within strata (seed 42) to include 130 of the 184 negated, 120 non-asserted and 150 asserted ones, and 200 sentences (100 containing a negation cue, 60 a hedge, attribution or condition cue, and 40 neither). For each fact they judged, with the stored labels visible, whether the stored negation flag and the stored modality were right; for each sentence, blind to the extractor, whether its main claim is negated and which of four modalities it carries (asserted, attributed, conditional, hypothetical). A language model was used only to reword the free-text notes that accompany one annotator's labels. 12 facts and 7 sentences that one annotator marked ambiguous or too thin to judge were excluded, leaving 388 facts and 193 sentences. Agreement between the annotators was $\kappa = 0.86$ for stored polarity, 0.50 for stored modality, 0.80 for sentence negation and 0.71 for sentence modality (prevalence-adjusted agreement, PABAK, 0.92, 0.81, 0.92 and 0.85). Table 7 reports precision per stored class and recall at sentence level.

The negation flag was wrong for more than half of the facts it marked negated: 43.9% of 123 stored-negated facts had correct polarity by annotator A (Wilson 95% interval 35–53%), 44.7% by B and 43.5% where both agree, against 99.6% of non-negated facts, and in 55 of the 69 wrong cases (80%) no negation cue occurs in the fact's own subject, verb or object (an automatic check), which indicates negation attached to a fact from

Measure	*n*	*Annotator A*	*Annotator B*	*A and B agree*
Polarity correct, stored negated	123	43.9% (54/123)	44.7% (55/123)	43.5% (47/108)
Polarity correct, stored not negated	265	99.6% (264/265)	100.0% (265/265)	100.0% (264/264)
Modality correct, stored asserted	255	93.3% (238/255)	92.5% (236/255)	96.6% (227/235)
Modality correct, stored attributed	89	78.7% (70/89)	71.9% (64/89)	80.0% (60/75)
Modality correct, stored conditional	12	100.0% (12/12)	100.0% (12/12)	100.0% (12/12)
Modality correct, stored hypothetical	32	93.8% (30/32)	96.9% (31/32)	100.0% (29/29)
Sentence negation, recall	21 / 25 / 19	81.0% (17/21)	80.0% (20/25)	78.9% (15/19)
Sentence negation, precision	56 / 56 / 49	30.4% (17/56)	35.7% (20/56)	30.6% (15/49)
Sentence non-asserted modality, recall	46 / 41 / 29	71.7% (33/46)	70.7% (29/41)	72.4% (21/29)
Sentence non-asserted modality, precision	46 / 46 / 33	71.7% (33/46)	63.0% (29/46)	63.6% (21/33)

Table 7: Extractor audit: precision of the stored negation flag and modality per stored class (facts), and negation and non-asserted-modality recall and precision at sentence level, per annotator and where both agree. Counts in brackets; n for sentence rows lists A / B / agreed. The sample is enriched for negated and non-asserted facts and for cue-bearing sentences, so precision is per class, not overall, and sentence precision is lowered by the enrichment.

another clause of the same sentence, not a missed cue. Sentence-level recall of negation was about 80% (81.0% by A, 95% interval 60–92%, on only 21 negated sentences), but sentence-level precision was about 30% in this cue-enriched sample, which is consistent with the same over-attachment. Modality labels were more accurate: asserted facts were correct 93.3% of the time by A (92.5% by B), attributed facts 78.7% (71.9%), and the few conditional and hypothetical facts were almost all correct, but annotators disagreed more on whether a stored modality was right ($\kappa = 0.50$), and A marked 38 stored modalities wrong, 3 of them without naming a different class; among the other 35 the most frequent corrections were attributed facts that were really asserted (17) and the reverse (7). The audit supports a limited claim: the fact layer's modality labels are mostly right for asserted facts, but its negation flag, which the system's grounding prompt asks the generator to respect, is not reliable at the fact level. It is a small, single-source sample with a language-model-worded note column and two annotators, and it does not measure the effect of these errors on answers; single-term SDH ablations and a negation- or hedge-sensitive question set remain future work.

6.9 Discussion

Keep structure, but not at the expense of prose. The system's rewrite of tables as extracted sentences was the weakest variant, the cheap structure-keeping baselines were never significantly ahead of tag-stripped chunks at the 2,048 ceiling, and on the fresh cohort every arm that kept tables and lists, including the cheap ones, was within a few points of every other. The evidence supports keeping native structure and does not separate G from whole-document Markdown, table-aware chunking or hybrid table-aware retrieval on documents of this length. Where the ceiling bound very tightly G was worse, and where it bound across several documents its lead over plain chunking did not survive correction. On full pages, where the ceiling bound throughout, G's structure-first packing with same-document suppression was clearly harmful, and a repaired variant that keeps prose facts (G2) restored parity with table-aware chunks on a fresh draw (Section 6.4). For short documents, passing the document as Markdown or using table-aware chunks serves as well as G; for long ones, table-aware chunks or the repaired G2 (at about twice their context) are clearly better than G. Rewriting prose as extracted facts did not help

either: the fact-only configuration was behind plain chunks in both settings where the two were compared, although the question-level interval of Section 6.7 includes zero and only one of the eight Holm-adjusted tests of Section 6.6 is below 0.05.

Answer formatting is a large effect on exact match and a small one on answer quality. The concise prompt raised exact match by 17 to 40 points (Table 5); the 2×2 follow-up shows that most of the gain is format, that the abstention wording is a separate factor whose effect depends on the generator, and that an equivalence judge sees far smaller format effects. For evaluation with small local generators, exact match should be accompanied by an equivalence measure, and prompt changes should be decomposed before a method comparison is read. With two generators, two prompt families and N = 100 it is not established that the size of these effects carries over to other models.

Explainability and practicality. Every packing decision can be traced to a similarity, a filter or the ceiling, and the interface exposes the same fields. The ceiling is derived from the host in the released code, extraction is rule-based, and all models are local. Only generation time was logged: in the main prompt experiment on the RTX 4060 host the median wall-clock time per answer was 0.22 to 0.93 s across the eight model, retrieval and prompt cells (90th percentiles 0.6 to 1.9 s; single calls up to 11.5 s, consistent with model loading), and the concise prompt lowered the median by 41 to 63%. Ingestion, retrieval and CPU-only timings were not measured, so no end-to-end speed claim is made, and no user study of auditability was run.

6.10 Limitations

(1) All cohorts come from Natural Questions (Wikipedia pages with short answers); all but the two long-document draws use MRQA's extracts of at most 800 words, on which the ceiling rarely binds; the long-document cohorts have a single generator, G2 was designed after the first of them and checked on one further draw from the same source, no multi-row-reasoning questions were tested, and the five-document corpora use random distractors. (2) The original table cohort is in-sample for G, its variants A–F reuse frozen predictions under a prompt wording that was not hash-recorded, and F2 is a reconstruction. (3) The non-inferiority margin of Section 6.3 was chosen after the fresh-cohort results (that of the long-document cohort was fixed before it was drawn), and the protocols are author-held, not externally registered. (4) The baselines omit cross-encoder reranking [Nogueira and Cho 2019] and row-serialised tables, the primary generator is code-specialised (a general-purpose Qwen generator and a Gemma generator, answering the same generator-independent contexts, leave G's position relative to the baselines unchanged), and each case is a single greedy generation. (5) Ingestion and retrieval latency, memory use and CPU-only operation were not measured (only GPU generation time was logged), and the token estimate is not calibrated against the generator's tokenizer. (6) The equivalence judge is a single local model checked by two author labellers on 100 outputs, and the pilot error audit has an author and a language model as raters. (7) The fact layer's contribution inside G is not isolated, its fact-only contexts are behind chunks, and its negation flag is unreliable; the fact-level audit showed the stored labels to the annotators, which may have anchored their judgements. (8) An earlier renderer doubled the negation marker in 9 of 200 stored F contexts and 4 of 100 prompt-experiment contexts and in none of G's; regenerating with the marker removed changed no exact-match outcome that could be tested, and the renderer has been fixed. The intent router is a set of English regular expressions, and the injection filter is a basic pattern list with a development default secret that must be replaced in production.

7 Conclusions and Future Work

This paper evaluated how much a structure-aware packing rule adds to local RAG over table-bearing pages and how much of a measured difference comes from answer format. Rewriting tables as extracted sentences cost most of the accuracy (14.0% against 49.5% exact match for native tables, in-sample), but against tag-stripped chunks the cheap structure-keeping baselines were never significantly ahead. On a fresh cohort G was not distinguishable from them, its non-inferiority rested on a post hoc margin, and it lost at a 256-token ceiling; on full Wikipedia pages it was the worst arm (22.0% EM against 33.0 to 41.5%) because same-document suppression dropped every prose fact, and a repaired rule (G2) gained 16.0 points and was non-inferior to table-aware chunking at about twice its context. A concise-answer prompt raised exact match everywhere, mainly through answer format, and an equivalence judge sees much smaller differences. Native structure should be kept but not at the expense of prose, and strict-match comparisons need an equivalence measure and controlled prompts.

Future work has five directions: (i) table-and-text benchmarks such as NQ-Tables [Herzig et al. 2021] and OTT-QA [Chen et al. 2021], a header-aware row-linearisation baseline, same-domain distractors and a cross-encoder reranker; (ii) testing the repaired rule G2 on other sources and refining its reference-list filter, row-level truncation for small ceilings, and a token estimate calibrated against the generator's tokenizer; (iii) repeated sampling and latency, memory and CPU-only measurements; (iv) a larger validation of the equivalence judge and the format–abstention decomposition with more generator families; and (v) fixing the negation-scope errors of the fact layer and testing its effect with single-term SDH ablations on a negation-sensitive question set.

A Prompt Templates and an Example Context

A.1 Templates of the main prompt experiment (Section 6.6). They were run with qwen2.5-coder:7b and qwen3.5:4b through the Ollama chat interface (thinking disabled; temperature 0.0, top-p 0.9, seed 42, 512 new tokens) and are given verbatim in the full manuscript in the repository: STANDARD (sha256 `dc584bb87b846fc60945b3c351ee8c405a980a1147aa63114eec67b3e2b3f4a5`) instructs the model to answer only from the supplied facts, to preserve negation and not to present attributed, conditional or hypothetical claims as established, and to say explicitly when the facts do not suffice; CONCISE (sha256 `3dc0137b82902e56840b6a27df6afeffe4b72cec027a06e08f69c24b734fc426`) keeps those rules but asks for only the shortest complete answer (an entity, name, date, number or exact phrase) and for the output "Insufficient information" instead.

Follow-up prompts (Section 6.6). Each changes only the abstention line. SHORT + EXPLICIT is CONCISE with the line "- If the provided facts do not contain enough information to answer the question, output "Insufficient information"." replaced by the STANDARD line "- If the provided facts do not contain enough information to answer the question, say so explicitly." (sha256 `055fe3eb608fa4438d7f602b20b514ede588a8d793c29ef80209630b41eaa524`); SENTENCE + TOKEN is STANDARD with the reverse replacement (sha256 `7758ebdb2bfa180152825ae285ef3ecf058a659b1d04119232d883783f84a209`). The format factor of Table 5 is therefore every other difference between STANDARD and CONCISE.

The pipeline's slightly different template wording, used for the fresh cohort, and an example of a packed context are given in the full manuscript in the repository (paper/research-paper/).

B Fact-Layer Specification and Implementation Notes

The fact layer's full specification (extraction rules, modality and negation labels, and the definitions and constants of the Semantic Density Heuristic) and the implementation notes (the negation-rendering defect summarised in Section 6.10, table header naming, extractor defects and the test suite) are given in the full manuscript in the repository; the fact layer is implemented in backend/src/precis/pipeline/svoa.py and sdh.py.

Author Contributions

Conceptualization, methodology, visualization, writing – original draft: Chandan D; Software, approval of final text: all authors; Investigation: Bhimsen; Formal analysis, validation, writing – review and editing: M P Jeevan Gowda; Data curation: Chandu S.

Data and Code Availability

The source code, experiment protocols, per-case records and analysis scripts are available at https://github.com/chandanxd/precis; a tagged release that reproduces each table and its archive DOI will be given here at submission [DOI to be inserted]. The Natural Questions data were taken from the MRQA 2019 shared-task release (https://github.com/mrqa/MRQA-Shared-Task-2019), development split, and, for the long-document cohort, from the full Natural Questions v1.0 development pages (Hugging Face release google-research-datasets/natural_questions). The generators and the judge of Table 1 were run through Ollama (the two main generators with Q4_K_M quantisation); the model digests and prompt-template hashes of the main prompt experiment are recorded in its configuration, and the protocols of the fresh-cohort, baseline, follow-up, generator-family, long-document and repaired-G experiments carry SHA-256 hashes and time stamps written before their model calls (author-held files, not an external registry); operational interruptions of the runs and the retries added are logged in each experiment's DEVIATIONS.md, and no completed case was regenerated. The annotation sheets, labels and scoring script of Sections 6.6 and 6.8 are in the repository, and the exact-match and containment rule used in every table is implemented once in scripts/scoring.py.

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